

GERSHON AYIEKO

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SUMMARY

- Analytical and detail-oriented Machine Learning and AI Engineer with hands-on experience in developing machine learning models, deep learning frameworks, AI automation, data engineering, and deploying scalable AI systems. Proficient in Python, SQL, TensorFlow, PyTorch, Scikit-learn, and cloud platforms like AWS for building predictive models, neural networks, and MLOps pipelines. Experienced in supervised/unsupervised learning, natural language processing, data preprocessing, model evaluation using metrics like accuracy, precision, recall, and F1-score, as well as transforming models into APIs for production integration. Passionate about leveraging advanced algorithms, mathematics, statistics, and problem-solving to create innovative AI solutions that optimize processes, drive business intelligence, and deliver measurable impact.

TECH STACK

- **Programming:** Python, SQL, JavaScript (for Apps Script)
- **Databases:** PostgreSQL, MySQL, Snowflake, MongoDB
- **BI & Visualization:** Power BI, Tableau, Looker Studio, Superset, Excel
- **ML & AI:** Predictive Modeling, Classification, Regression, Model Deployment
- **Automation:** n8n, Google Apps Script, API Integrations
- **Tools:** Git, Jira, Google Workspace, ETL Pipelines
- **Cybersecurity & Data Governance:** Data Protection, Security Fundamentals

SOFT SKILLS

- Deep analytical mind
- Problem-solving
- Curiosity
- Reliability
- Avid learner

EDUCATION

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| 1. Cooperative University of Kenya
BSc in Data Science | September 2022 - August 2026 |
| 2. Strathmore University - Data Privacy & Protection | September 2024 - October 2024 |
| 3. CyberShujaa Program
Cyber Security | June 2024 - August 2024 |
| 4. Microsoft Learning & LinkedIn
Data Analysis | April 2024- May 2024 |

PROJECTS

Traffic Speed Detection – Computer Vision (OpenCV)

A real-time vehicle speed detection system built using Python and OpenCV. The project tracks moving vehicles frame-by-frame, calculates motion displacement over time, and estimates speed based on distance covered in the video. It includes object detection, tracking logic, ROI calibration, and speed flagging for over-speeding vehicles.

[Link](#)

Technology: Python, OpenCV, NumPy

Judge Scoring System – LAMP Stack

A full-stack judging and score aggregation system built using the LAMP stack (Linux, Apache, MySQL, PHP). The system allows judges to enter scores in real time, manages user roles (admin, judge), validates entries, and displays final results on a live leaderboard. Designed for events, competitions, and talent shows.[link](#)

Technology: PHP, MySQL, Apache, HTML/CSS

BizOps Maintenance Track App(Deployed)

An internal BizOps tool designed to help teams track equipment maintenance, repairs, and operational tasks.

The app integrates with Google Sheets API for real-time updates, sends notifications for overdue tasks, and maintains an organized view of asset health and pending work. Improves operational efficiency and reduces manual tracking.

Technology: Typescript,Tailwind, Google Sheets API

UniLibrary Management System

Built a library management system supporting inventory, user management, borrowing, and reporting. [link](#)

Loan Default Prediction Model – ML Pipeline (In Progress)

An end-to-end machine learning pipeline built to predict loan default probability using credit history, demographics, and transactional data. Includes feature engineering, data balancing, hyperparameter tuning, and model evaluation using classification metrics.

Technology: Python, Scikit-learn, Pandas, NumPy

Product Recommendation System – AI Engineering (In Progress)

A hybrid recommendation engine combining collaborative filtering and content-based filtering. The system analyzes user behavior, product metadata, and similarity scores to generate personalized product recommendations. Designed to be deployed via an API layer.

Technology: Python, Scikit-learn, Pandas, FastAPI

LinkedIn AI Content Creator Agent

An AI automation agent that generates LinkedIn posts, repurposes content, suggests topics, and schedules posts automatically. Uses an LLM pipeline to process user input, generate text, and optimize tone for engagement.

Technology: OpenAI API / LLM, N8N

Customer Support AI Agent

A conversational AI agent designed to handle customer queries, search FAQs, retrieve information from a knowledge base, and escalate complex issues. Uses NLP to understand user intent and respond accurately.

Technology: N8N, LLM API, RAG , Pinecone

RAG Pipeline & Custom Chatbot

A Retrieval-Augmented Generation (RAG) system that combines vector embeddings, semantic search, and an LLM to answer questions based on uploaded documents. Includes embedding generation, vector store indexing, and a custom chatbot interface.

AI Lead Qualification Agent (In Progress)

An AI agent that analyzes incoming leads, scores them using defined criteria, classifies intent, and auto-updates CRM entries. Helps sales teams focus on high-quality leads while automating repetitive assessment tasks.

Technology: LLM API, N8N

Additional Skills

- ETL Development & Pipelines
- Machine Learning & Predictive Modeling
- Data Engineering & Automation
- Data Governance & Analytics
- IT Support & Troubleshooting
- Stakeholder Communication & Problem-Solving

WORK EXPERIENCE

Intern-Data & IT Department

Watervale Investment Limited (Moko Furniture & Living), | June 2025 – September 2025

- Gained hands-on experience in Data science, data analytics and IT support during a 10-week internship.
- Performed data cleaning, validation, and transformation using Excel, SQL, and Python (Pandas).
- Designed and deployed interactive dashboards in Power BI, Superset, and Looker Studio for sales, marketing, and inventory.
- Built and maintained MySQL and Snowflake queries for extracting and analyzing business-critical datasets.
- Participated in the Inventory Discrepancy Project: anomaly detection, automated reporting, and insights for management.
- Developed a maintenance tracking web application(BIZOP MAINTENANCE APP) integrated with Google Sheets API.
- Automated workflows using Google Apps Script and n8n, including email alert systems for scheduled maintenance.
- Supported Jira Analytics: extracted and visualized project/task performance data.
- Contributed to the Cradle Variance Project using advanced SQL for variance tracking.
- Provided IT support (system upgrades, troubleshooting, user support, and device maintenance).
- Strengthened teamwork, communication, problemsolving, and reporting skills.

Data Analytics Job Simulation

Deloitte| May 2025 – June 2025

- Completed a simulation focused on analytics, forensic technology, and business intelligence tasks.
- Built dashboards using Tableau and Looker Studio.
- Used Excel for data classification and deriving business insights.

REFEREES

1. Vincent Agan - Head of IT, Watervale Investment Limited

Email : agan@watervale.co.ke

2. Christopher Khaemba- Head of IT infrastructure, Watervale Investment Limited

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